

Tut Test 3

The Bode plot on the back of this page shows the frequency response of an unknown system.

- a) What is the order of the system? Explain your answer with reference to the plot. [2 marks]
- b) Using features of the plot, explain why it would be acceptable to model the system as second-order. For what range of frequencies would the second-order model hold? (You may give a loose range in terms of the nearest decade.)
[3 marks]
- c) Use the plot to estimate a second-order transfer function for the system. [5 marks]

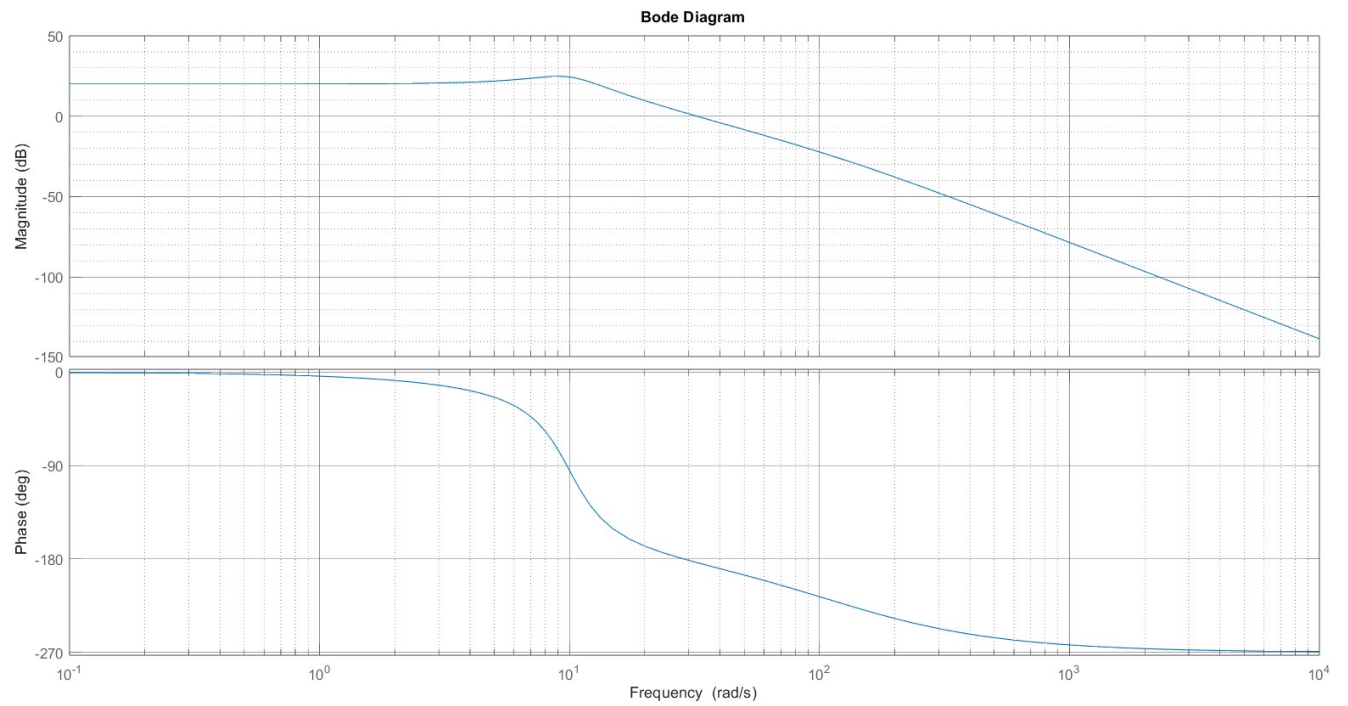


Figure 1: Bode plot for frequencies up to 10^4 rad/s.

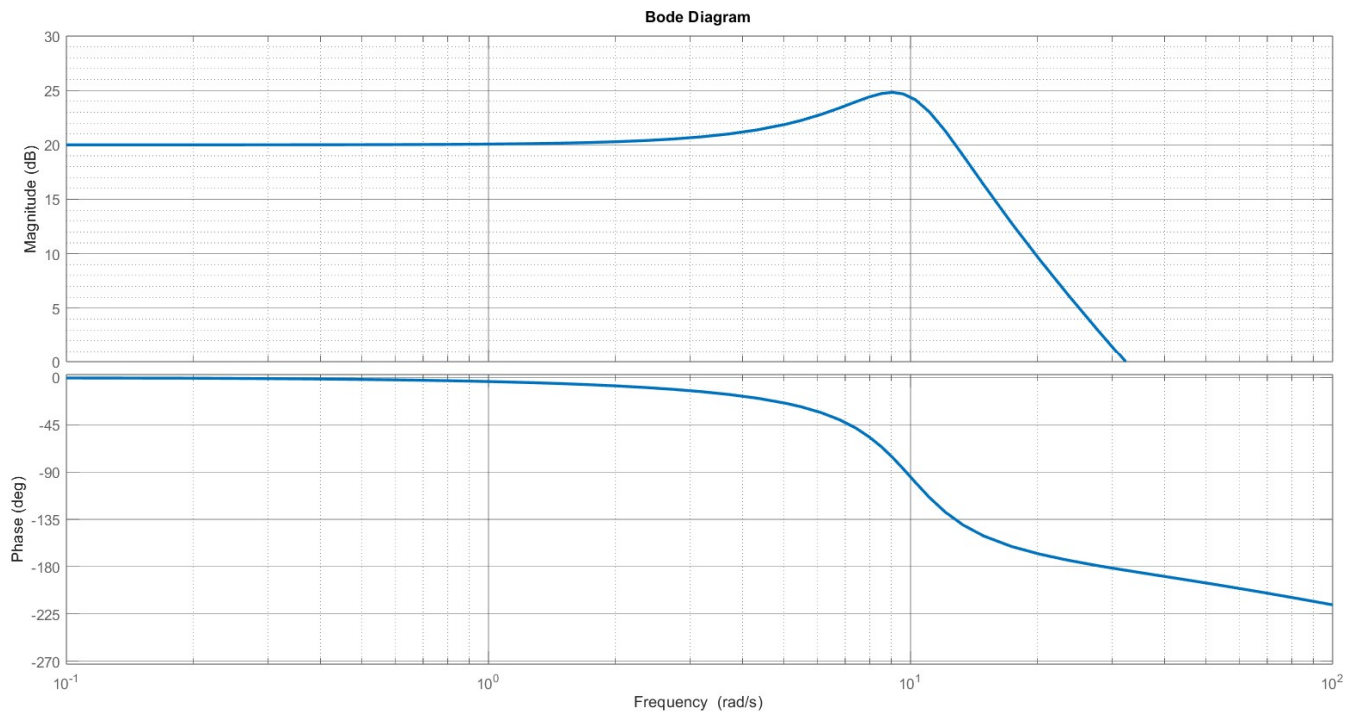


Figure 2: Zoomed in Bode plot for same system

Solution:

- a) The system must have at least three poles as the phase response tends to -270 degrees at high frequencies (-90 degrees per pole) OR because the roll-off at high frequencies is 60 dB per decade (20 dB of roll-off per pole).

[1 mark for number, 1 mark for explanation. No marks if not explained/ incorrectly explained.]

- b) Based on the resonant peak, the system appears to have a pair of complex poles, with a faster real pole. The location of the peak puts the natural frequency of the complex poles in the order of 10 rad/s, while the real pole has a corner frequency upwards of 100 rad/s based on the transition to 60 dB roll-off. Because the real pole is therefore more than 10 times faster than the complex poles, it can be neglected in the model. The second-order model will therefore hold for frequencies up to the 100s of rad/s.

[2 marks for justification of 2nd order model, 1 mark for range of frequencies and explanation.]

- c) A second-order transfer function has the form $G(s) = \frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$.

The low-frequency magnitude of the Bode plot is 20 dB, which gives a gain $A = 10$.

The phase crosses -90 degrees at 10 rad/s, indicating that this is the natural frequency, ω_n .

The magnitude at the natural frequency is around 24 dB, or 15.85.

This means $15.85 = \frac{A}{2\zeta}$, $\therefore \zeta = \frac{10}{2(15.85)} = 0.3$

This gives the transfer function $G(s) = \frac{1000}{s^2 + 6s + 1}$

You can also use the magnitude of the resonant peak, and resonant frequency to find these values.

[1 mark for gain, 2 marks for natural frequency, 2 marks for damping ratio]